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COURSE SYLLABUS IN AEE 321 – MICROPROCESSOR SYSTEMS

PART I. UNIVERSITY INFORMATION

VISION	Premier research university transforming communities in the ASEAN and beyond.
MISSION	USeP shall contribute to inclusive growth and sustainable development through continual innovation in proactive academic programs, impactful research for utilization, economically empowering community services, and sustainable resource management.
GOALS	● Internationalized University programs and services. ● Innovation and RDE-nurturing environment. ● Proactive, globally competitive, and service-oriented graduates. ● Transformed vulnerable and disadvantaged groups and communities. ● Mobilized and optimized resources for sustainable development.
EDUCATIONAL PHILOSOPHY	
Nature of Education in the University	The University of Southeastern Philippines is a state university that contributes to the development of human capital and to fostering innovations to transform communities in Mindanao, the Filipino nation, and the world. As a community of scholars, we aim to develop life-long learners who, guided by conscience, actively engage in the pursuit of generating and extending knowledge through research, extension, and responsible professional practice.
Principles for Teaching and Learning in the University	In all our educational endeavors, we strive to be continually guided by the following principles: ● Teaching and learning is based on science and evidence stemming from experience and reflections on the teaching practice. ● A conducive learning environment is founded on integrity and mutual respect, where learning is a co-construction among scholars. ● The learner is an active participant in the learning process to develop oneself and others' potential. ● A transformative learning experience is grounded in critical and reflective thinking and provides opportunities for practice, improvement, and action.
INSTITUTIONAL GRADUATE ATTRIBUTES	
IGA1: LEADERSHIP SKILLS	Creates and inspires positive changes in the organization; exercises responsibility with integrity and accountability in the practice of one's profession or vocation.
IGA2: CRITICAL AND ANALYTICAL THINKING SKILLS	Demonstrates creativity, innovativeness, and intellectual curiosity in optimizing available resources to develop new knowledge, methods, processes, systems, and value-added technologies.
IGA3: SERVICE ORIENTED	Demonstrates concern for others, practices professional ethics, honesty, and exemplifies socio-cultural, environmental concern, and sustainability.
IGA4: LIFELONG LEARNING	Demonstrates enthusiasm and passion for continuous personal and professional development.
IGA5: PROFESSIONAL COMPETENCE	Demonstrates proficiency and flexibility in the area of specialization and in conveying information in accordance with global standards

PART II. PROGRAM INFORMATION

Program Name: Bachelor of Science in Electrical Engineering

PROGRAM GRADUATE OUTCOMES			
PGO	**Description**	**Performance Indicator**	**IGA Aligned to**
PGO a	Apply knowledge of mathematics and sciences to solve engineering problems	a.1. Apply concepts in mathematics to solve engineering problems.	IGA2
PGO e	Identify, formulate, and solve engineering problems	e.2. Apply concepts, theories, and principles in solving engineering problem(s)	IGA5

PART III. DOCUMENT INFORMATION

Version Number	:	2026–1.0
Version Note	:	✓ New ☐ Minor Revision ☐ Major Revision ☐ No Revision
Describe Revision *(if applicable)*	:	
Semester	:	Second Semester
Academic Year	:	2025-2026

PART IV. FACULTY INFORMATION

Faculty Name	:	ELWIN JAY J. TANO
Email Address	:	ejjtano0120241110@usep.edu.ph
Contact Number	:	09262995868
Office	:	College of Engineering
Consultation Hours	:	Monday-Friday, 8:00 AM – 05:00 PM (excluding lunch break)

PART V. COURSE INFORMATION

Course Code	:	AEE 321
Course Title	:	Microprocessor Systems
Pre-requisite	:	AEE 313 – LOGIC CIRCUITS DESIGN AND SWITCHING THEORY AEE 313L – LOGIC CIRCUITS DESIGN AND SWITCHING THEORY
Credit	:	2.0 Unit - Lecture
Level	:	Third Year

SDG Alignment:

SDG	Description of Alignment
9: Industry, Innovation, and Infrastructure	Microprocessors and microcontrollers are foundation technologies for industrial automation, smart devices, and digital infrastructure. Arduino-based system design fosters innovation and practical engineering skills.
11: Sustainable Cities and Communities	Microcontroller-based systems are key enablers of smart city technologies, including traffic control, environmental monitoring, energy-efficient lighting, and public safety systems.

Course Description:

The course includes history and evolution, principles, and applications of microprocessors. The focus is on the basic understanding of the architectural design, functional parts, operations, function, and programming. It also covers the study of various types of microprocessors and the fundamental concepts of microcontrollers.

Workload of Students:

With **18 weeks** at **2 contact hours** per week (**36 contact hours total**), the expected student workload is as follows:

- Total of 4 quizzes (2 for both Midterm and Final Term)
- Post-discussion oral recitation or group activity
- Midterm and Final Examinations

In instances where face-to-face sessions are impractical, simulated experiments or their equivalent will be provided.

Course Learning Outcomes (CLOs)

On the completion of the course, the student is expected to be able to do the following:

Code	Course Learning Outcomes	PGO Aligned to
CLO1	Analyze and compute the operations of ALU, CU, and registers in microprocessors/microcontrollers using mathematical methods.	PGO a
CLO2	Design and troubleshoot a microcontroller based I/O interfacing systems involving data movement and control.	PGO e

PART VI. ASSESSMENT ACTIVITIES

Formative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
FA1	Oral Recitation	Oral recitations require students to explain and analyze the functional operations of microprocessors and microcontrollers—such as ALU processing, control unit sequencing, and instruction execution—using Arduino-based examples. Emphasis is placed on articulating how these internal operations support the development of reliable and innovative embedded systems used in modern infrastructure and smart community applications.	CLO1
FA2	Group Activity	Group activities engage students in the collaborative design, implementation, and troubleshooting of Arduino-based microcontroller systems involving data movement, control logic, and I/O interfacing. Through teamwork and hands-on system development, students simulate real-world engineering practices while creating embedded solutions relevant to intelligent infrastructure and sustainable community technologies.	CLO2

Summative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
SA1	Quizzes	Quizzes assess students' ability to analyze and solve problems related to microprocessor/microcontroller structures, instruction sets, and functional operations. These assessments reinforce the foundational knowledge required to develop dependable microcontroller-based systems that underpin innovative industrial applications and efficient, technology-driven urban environments.	CLO1, CLO2
SA2	Term Examination	The term examination comprehensively evaluates students' understanding of microprocessor and microcontroller principles through problem-solving, analytical, and design-oriented questions centered on Arduino-based systems. Students are expected to integrate theoretical concepts and practical reasoning to demonstrate how embedded systems contribute to innovative technological infrastructure and sustainable, smart community solutions.	CLO1, CLO2

PART VII. MEASUREMENT SYSTEM

Assessment Activity: FA1: Oral Recitation

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Conceptual Understanding	Excellent understanding; fully accurate and thorough explanation.	Good understanding; mostly correct.	Partial understanding; minor errors.	Basic understanding; some correct points.	Very limited understanding; mostly incorrect.	No understanding; cannot explain topic.
Application & Examples	Fully correct and relevant application to Arduino code/hardware/examples.	Mostly correct application.	Partial application; minor errors.	Some application; partially correct.	Very limited or incorrect application.	No application
Clarity & Communication	Very clear, confident, logical, and precise explanation.	Mostly correct application.	Partial clarity; understandable but not structured.	Some clarity; partially organized.	Very poor; hard to understand.	Incoherent; cannot communicate.

Assessment Activity: FA2: Group Activity

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Task Completion & Accuracy	Fully completed; functional, accurate, meets objective	Completed correctly; mostly functional	Mostly completed; minor errors	Partially completed; some correct elements.	Very minimal completion; mostly incorrect.	Not completed
Problem-Solving & Troubleshooting	Excellent problem identification, solution, and reasoning	Mostly correct identification and solutions.	Partial identification and solutions.	Identifies some issues; solutions are partially correct.	Very limited or incorrect solutions.	No-problem solving
Collaboration & Documentation	Excellent teamwork, complete, and clear documentation/report/presentation.	Good teamwork; mostly clear documentation.	Partial teamwork; documentation acceptable	Some teamwork documentation acceptable.	Very limited participation or poor documentation.	No teamwork or documentation

Assessment Activity: SA1/SA2: Problem Solving Rubric

Area to Assess	Excellent (100%)	Very Good (80%)	Satisfactory (60%)	Needs Improvement (40%)	Unsatisfactory (20%)	Unacceptable (0%)
Conceptual Analysis	Excellent understanding; fully accurate and thorough analysis.	Good understanding; mostly correct.	Partial understanding; minor errors.	Basic understanding; minor errors.	Very limited understanding; mostly incorrect.	No understanding; cannot analyze problem.
Problem-Solving & Application	Fully correct, efficient, and logically sound solution.	Mostly correct solution; minor issues.	Adequate solution; partially correct.	Partially correct solution; minor errors.	Very limited solution; mostly incorrect.	No attempt or completely incorrect.
Accuracy & Logical Reasoning	Fully logical, stepwise, and accurate.	Mostly logical and accurate.	Reasonable logic; some minor errors.	Some logical steps; partial correctness.	Very limited reasoning; mostly incorrect.	Completely illogical or wrong.

Assessment Activity: SA1/SA2: Short Essay Rubric

Area to Assess	Excellent (100%)	Very Good (80%)	Satisfactory (60%)	Needs Improvement (40%)	Unsatisfactory (20%)	Unacceptable (0%)
Conceptual Understanding	Excellent understanding; fully accurate, thorough, and insightful.	Good understanding; mostly correct.	Partial understanding; minor errors.	Basic understanding; some correct points.	Very limited understanding; mostly incorrect.	Shows no understanding;

						explanation missing or incorrect.
Application & Examples	Fully correct, relevant, and well-illustrated examples.	Mostly correct application.	Partial application; minor errors.	Some application; partially correct.	Very limited or incorrect application.	No application
Clarity, Organization, & Reasoning	Excellent clarity, highly logical, well-structured, and precise.	Good clarity; mostly logical and organized.	Partial clarity; understandable but not structured.	Some clarity; partially organized.	Very poor; hard to understand.	Completely incoherent

PART VIII. GRADING SYSTEM AND ASSESSMENT PLAN:

The final grade in this course will be composed of the following items and their weights in the final grade computation:

Assessment Item	Percentage of Final Grade	CLO1	CLO2
SA1	40%	Objective / Concept-type Items Score / Problem Solving Items Score (with Rubric in Problem Solving) / Short Essay Items Score (with Rubric in Short Essay)	Objective / Concept-type Items Score / Problem Solving Items Score (with Rubric in Problem Solving) / Short Essay Items Score (with Rubric in Short Essay)
SA2	60%	Objective / Concept-type Items Score / Problem Solving Items Score (with Rubric in Problem Solving) / Short Essay Items Score (with Rubric in Short Essay)	Objective / Concept-type Items Score / Problem Solving Items Score (with Rubric in Problem Solving) / Short Essay Items Score (with Rubric in Short Essay)
TOTAL	**100%**		

Transmutation	:	Base-50
Passing Grade	:	3.00
Passing Grade Conditions	:	Students shall take all assessments and examinations and must have an attendance of not less than 80% in the class. In case of lacking output or examination, a grade of INC (Incomplete) shall be given in accordance with the existing rules and regulations reflected in the student handbook. In case of incurring absences of more than 20% of the total expected number of attendances, a grade of UW (Unauthorized Withdrawal) will be reflected in the grade sheet.

Grade Equivalence Table:

< 75 = 5.00	75-76 = 3.00	77-79 = 2.75	80-82 = 2.50	83-85 = 2.25
86-88 = 2.00	89-91 = 1.75	92-94 = 1.50	95-97 = 1.25	98-100 = 1.00

PART IX. LEARNING PLAN

In order to achieve the outcomes of this course, learners will go through this learning plan:

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
• Recognize the university VMGO & Institutional Graduate Attributes. • Appreciates the significance of PEOs and PGOs of the programs. • Recognize the rules and regulations of the course and the grading system.		1 hour	• Course Orientation • VMGO & IGA of the University • PEOs & PGOs of the program. • Rules and regulations of the course & grading system • Presentation of the course outline	• On-site or online meeting	• UseP Student Handbook (2016 edition)	• N/A
• Analyze the interactions between structural components of a microprocessor/microcontroller to explain how they work together to perform tasks. (CDIO Skills: 1.1, 1.2, 2.1)	CLO1	4 hours	• **STRUCTURAL COMPONENTS OF MICROPROCESSOR/MICROCONTROLLER** o Internal CPU Interconnection o Arithmetic Logic Unit o Control Unit o Registers	• Interactive demonstration and discussions. • Oral Recitation	• Ref01 • Ref04-08	• FA1 – Oral Recitation • FA2 – Group Activity • SA1 – Quiz
• Evaluate the fetch-decode-execute cycle to determine the timing and sequencing of instructions in a microprocessor. (CDIO Skills: 1.2, 2.1, 2.3)	CLO1	5 hours	• **FETCH-DECODE-EXECUTE CYCLE** o Fetch Stage o Decode Stage o Execute Stage o Store Stage o Execution Time and Instruction Time	• Interactive demonstration and discussions. • Oral Recitation	• Ref01 • Ref05 • Ref07-08	
• Develop a program that implements data movement, processing, and control using a microcontroller to meet specific functional requirements. (CDIO Skills: 1.2, 2.1, 3.1)	CLO2	6 hours	• **FUNCTIONAL OPERATIONS OF MICROPROCESSORS/MICROCONTROLLER** o Data Movement o Data Processing o Control o Data Storage	• Interactive demonstration and discussions. • Oral Recitation	• Ref01 • Ref05-08	• FA1 – Oral Recitation • FA2 – Group Activity • SA1 – Quiz
• Pass the midterm examination	CLO1 CLO2	2 hours	**MIDTERM EXAMINATION**	• **Conduct of Midterm Examination**		• **SA2 – Term Examination**
• Apply the microcontroller's instruction set to perform arithmetic, logical, and data transfer operations. (CDIO Skills: 1.2, 2.1, 3	CLO1	8 hours	• **INSTRUCTION SET** o ATMega328P Opcodes o Instruction Cycle o Instruction Categories: ▪ Data transfer ▪ Arithmetic/Logic ▪ Control Flow ▪ Bit Manipulation ▪ Stack/Subroutine ▪ Interrupt Control ▪ Timing/Performance	• Interactive demonstration and discussions. • Oral Recitation	• Ref01-08	• FA1 – Oral Recitation • FA2 – Group Activity • SA1 – Quiz

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
• Design a safe and functional IoT system for home automation using ESP32 and relay modules (CDIO Skills: 4.4, 2.1, 2.3)	CLO2	8 hours	• **I/O INTERFACING** ○ Interfacing of Input/Output Devices ○ Interface Devices ○ Time-Based I/O ○ Handshaking	• Interactive demonstration and discussions. • Oral Recitation	• Ref01-08	• FA1 – Oral Recitation • FA2 – Group Activity • SA1 – Quiz
• Pass the final examination	CLO1 CLO2	2 hours	**FINAL EXAMINATION**	• **Conduct of Final Examination**		• **SA2 – Term Examination**

PART X. REFERENCES/ARTICLES TO READ

Code	Reference/Article
Ref01	Hughes, J.M. (2016). *Arduino: A Technical Reference*. O'Reilly Media, Inc.
Ref02	Norris, D. (2015). *The Internet of Things: Do-It-Yourself Projects with Arduino, Raspberry Pi, and BeagleBone Black*. McGraw-Hill Education
Ref03	Guptha, M.A. (2021). *Internet of Things & Its Applications*. Malla Reddy College of Engineering & Technology
Ref04	Monk, S. (2016). *Programming Arduino™: Getting Started with Sketches (2nd ed.)*. McGraw-Hill Education
Ref05	Blum, J. (2013). *Exploring Arduino: Tools and Techniques for Engineering Wizardry*. John Wiley & Sons, Inc.
Ref06	Margolis, M. (2011). *Arduino Cookbook*. O'Reilly Media, Inc.
Ref07	Nussey, J. (2018). *Arduino for dummies (2nd ed.)*
Ref08	Atmel. (2015). *Atmega328P Datasheet*

PART XI. CLASSROOM POLICIES

1. **Attendance**. Classroom attendance is an integral part of the educational experience; therefore, it is the student's responsibility to attend class. Attendance will be attended at every meeting.
2. **Behavior and Decorum**. Respectful, inclusive, and professional behavior is always expected. The use of polite and non-discriminatory language is strictly observed in line with the University's commitment to a gender-fair and inclusive learning environment, and adherence to RA 11313, also known as the Safe Spaces Act.
3. **Adherence to Safety Protocols**. All equipment and materials must be handled with care; any misuse or negligence will result in reduction in the grade of laboratory activity. Students are expected to follow mandatory safety protocols, especially proper handling of electronic devices and checking the actual circuit before energizing.
4. **Submission of Coursework**. Extensions and late submissions of requirements will only be considered if done in writing to the lecturer at least two days before the due date. Generally, no extension will be given for the assignments. Please contact your subject adviser if you encounter any problems.
   Note: (Submission / Resubmission):
   - *Should work be submitted after the due date without written approval for an extension, a penalty of 5% deduction from the grade will be incurred for every 2 days it is late.*
   - *Should the work be assessed as being close to a PASS grade, but the standard has not yet been fully met, the lecturer may offer the opportunity to resubmit.*
5. **Use of AI Tools**. The responsible use of generative AI tools (e.g., ChatGPT, Grammarly, Quillbot) is permitted only when explicitly allowed by the instructor for specific tasks. All AI-assisted work must be properly cited or declared. Utilizing generative AI tools for generating code, especially during laboratory activities is strictly prohibited. Any group who will be caught will be given half of the points on that activity.
6. **Academic Honesty**. All USePians are bound to follow the rules and regulations stipulated in the student handbook of the University and the (College/School). All University policies regarding academic honesty apply to this course. Academic dishonesty includes, but is not limited to, cheating in any form, plagiarism, and similar offenses.
   - Cheating is a serious offense. If you cheat on any assignment or examination, the minimum penalty is a grade of zero for that work. Your conduct may also be referred to the Office of Student Affairs and Services (OSAS) for appropriate sanctions under university policy.
   - Students are encouraged to work together in exchanging ideas and discussing their assignments in general. However, it is also expected that the work of analyzing data and writing reports will be done in their respective groups. If reports or data have been copied from another student, both students will receive a mark of zero on that examination, test, report, or assignment.
   - Submitting AI-generated content as original work without disclosure is considered academic dishonesty. The acceptable level of matched content using university-accepted plagiarism detection software is 15%. Any paper exceeding this threshold may be subject to rejection or revision.
7. **Conduct of Quizzes and Examinations.** Every quiz and exam, students must keep the jacket of their calculators, smartphones, and other things irrelevant to the quiz or exam in their respective bags. If the student caught with any of those objects during the quiz and/or examination proper, it would serve as cheating and dealt accordingly in accordance with the policies imposed by the latter. Instructions must be observed with utmost compliance. Unable to do so will lead to consequences (e.g., deduction of points or answers will not be recognized). During the solution part, rounding up is not recommended as it will accumulate rounding-off errors to the final answer. Final answers must be in three decimal places. Math plate is observed to be used during quizzes and bluebook for term examinations. In case the former is unavailable, bluebook can be used as a substitute.
8. **Use of Calculators.** Allowed calculators must follow the Professional Regulatory Commission (PRC) recommendations, see https://www.prc.gov.ph/node/6984 for the full list of non-programmable calculators allowed by the Commission.
9. **Cleanliness and Energy Conservation**. After each class session, ensure that everything is in order and all plugged devices and lights are turned off before leaving the classroom.

Prepared by:	Reviewed by:	Approved:
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